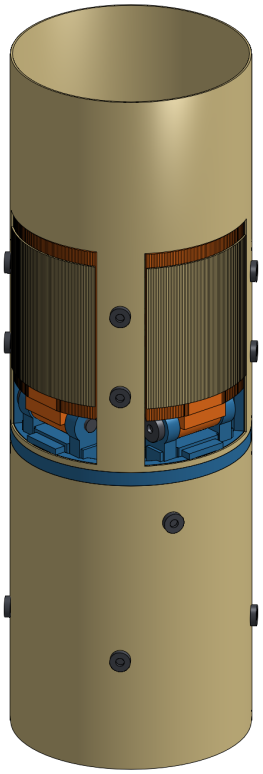
V2 – Compact head assembly guide

Same mechanism, hardware and fins as V1, but the fins live in a smaller-diameter head that slides *inside* the upper BT80 tube, and the whole latch stack (hook, crown arms, top bearing, knob) now sits **inside the fin height**: the fin top is only 4.8 mm below the head top. You cut four slits in the tube; each cut-out piece is glued onto its fin and rides with it.

Onshape: ARC BT80 Airbrake - V1 Screwed + V2 Compact (Claude build). Elements: **V2-A** closed, **V2-B** fully open, **ASM-V2** assembly, Feature Studio parameter *V2 compact head* = on.



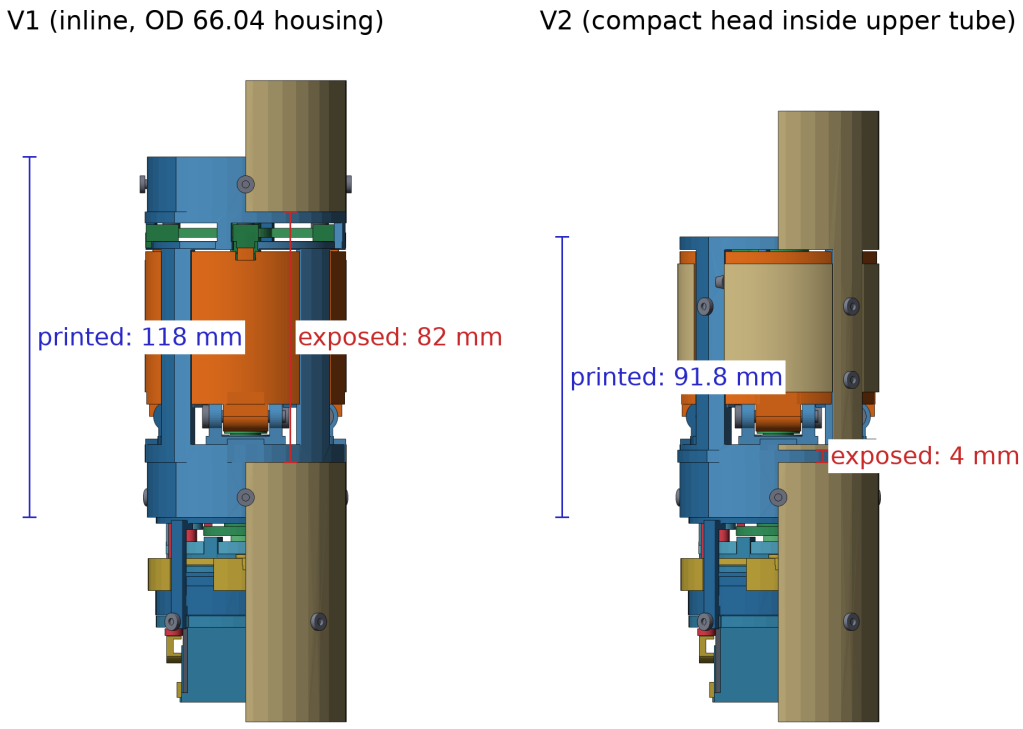
V2 closed. The upper tube covers the fin zone; only a 4 mm stop ring of the airbrake is exposed between the tubes.

1 The idea and the numbers

Your sketches: slit the upper BT80 tube around the fins, glue the cut-outs to the fins, keep the sleeve fit at the bottom, put supports in the solid strips between the fins. Your fin-position drawing: the fin goes up to the top rim and everything under it moves up with it, so the whole piece is shorter than V1.

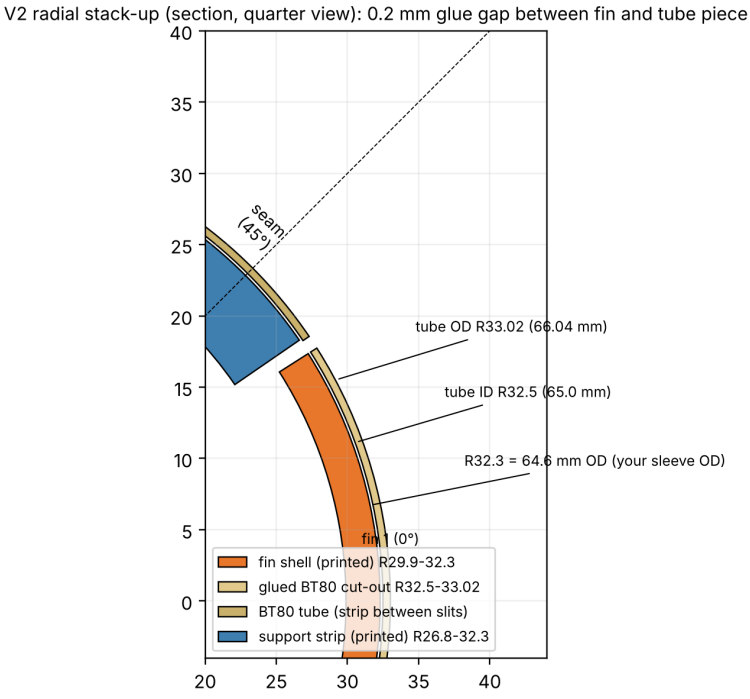
Item	V1	V2 (22 mm drop)
Fin shell radii (printed)	R30.62 – 33.02	R29.9 – 32.3 (64.6 mm OD = your sleeve OD)
Outer skin	printed, flush with the tube (66.04 mm)	glued BT80 cut-out R32.5 – 33.02 (0.52 mm wall); 0.2 mm glue gap to the fin
Head OD	66.04 mm (flush)	64.6 mm inside the tube (tube ID 65.0 mm)
Fin catch	tab above the fin tip, hook outside the skin (z 51–63)	inward lug on the fin (plate z 47–50, inner wall z 41.6–50) hooked from inside by a finger on the catch cap (z 40.8–46)
Latch stack on the rotor	crown arms z 58.6–61.6, top bearing 68–73, knob 74–78	all 22 mm lower: arms 36.6–39.6, top bearing 46–51, knob 52–56
Upper sleeve	18 mm, z 67..85	removed; 4.2 mm top ring z 54.6..58.8
Supports	none (thin 2.8 mm pillars)	4 thick strips between the fins R27.4–32.3 (two pieces, clear of the cam band z 18.5..29.5) + the 4 under-fin stop blocks R27–32.3
Lower sleeve	18 mm, OD 64.6 mm	unchanged (sleeve fit)
Fin top to head top	28 mm	4.8 mm
Exposed airbrake between tubes	82 mm (z –15..67)	**4 mm** stop ring (z –15..–11)
Printed housing incl. sleeve	118 mm (z –33..85)	91.8 mm (z –33..58.8)
Whole printed assembly incl. cradle	178.5 mm	**152.3 mm** (26.2 mm shorter than V1)

Item	V1	V2 (22 mm drop)
Cam peak radius	29.0	28.6 (keeps 0.9 mm to the pillars)



V1 vs V2 side by side (tubes cut in half for the view). The exposed band drops from 82 mm to 4 mm and the printed housing from 118 to 91.8 mm.

Why supports are still needed: a flush butt joint between two tubes has no load path and nothing to retain it. In V1 the pillar wall and the 18 mm glue sleeves carry that; in V2 the thin wall is gone, so the thick strips (screwed to the tube) and the sleeve fit take over.



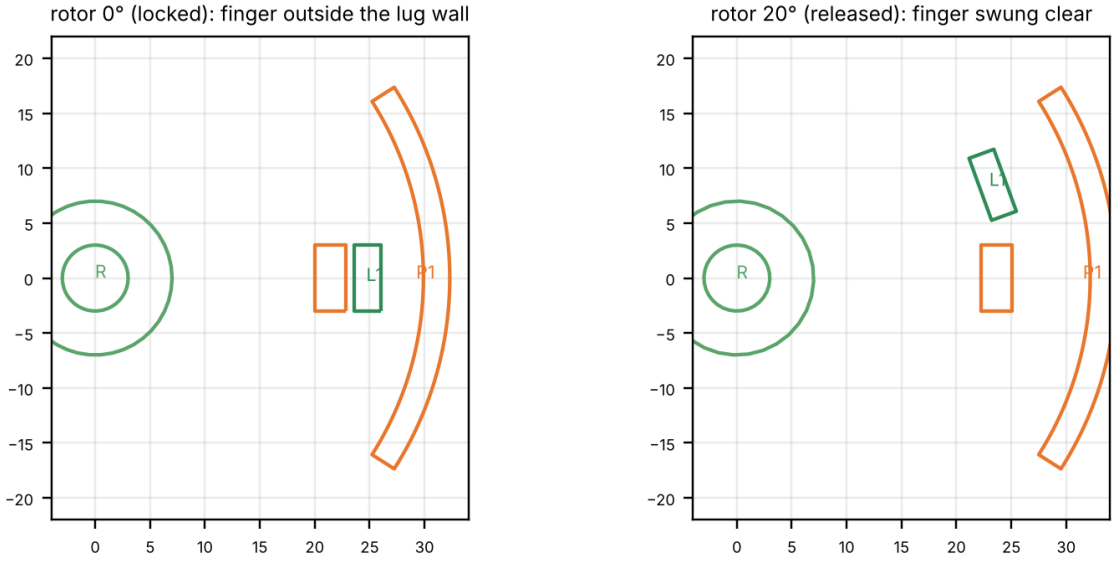
Radial stack at a seam strip (z=36): fin, 0.2 mm glue gap, glued tube piece, support strip.

2 The new latch (what changed to get the fin to the rim)

In V1 the fin tip carried a tab that a hook on the rotor grabbed from outside, above the fin. That hook swings sideways through the fin's own window, so it cannot sit inside the fin height. In V2 the catch is turned inside out: the fin carries a small **lug** that reaches inward (plate at z 47–50 and an inner wall to z 41.6). The catch cap on each rotor arm carries a **finger** that stands just outside that wall (0.8 mm gap). The fin cannot move out until the rotor swings the finger away; the finger runs at R23.6–26, inside the support strips (R27.4+), so it can sweep past them.

- Hold: at rotor 0° the fin can move about 1° (0.8 mm at the lug) and is blocked from 2° (interference scan of the CAD).
- Release: with the fin held at 3°, the fin is free once the rotor is at 16° or more (on the final model: still held at 14°, free at 16°).
- Timing: the cams first touch the fins between 22° and 26° of rotor travel (on the final model: no contact at 22°, contact at 26°), so the fins are released before they are pushed.
- Finger and wall are 6 mm wide, finger 2.4 mm thick: the latch is a printed hook, so it is thinner than the V1 hook was strong; run it empty first.

Section at z = 43.5 mm, looking down. Orange: fin lug wall; green: cap finger and rotor arm



Plan section at z=43.5: locked (left) and released (right). Orange: fin lug wall; green: finger on the catch cap and the rotor arm.

3 Print list

STLs: `onshape/exports/v2/print`. Same files as V1 except the housing halves, fins (lug) and catch caps (finger). Cut-outs TP1–TP4 are **not printed**.

File	Qty	Part	Notes
C1_housing_half_1_x1	1	C1 housing half (fins 1 and 4, pilot holes of the seam, 4 thick strips)	Pegs/sockets align it with C2. Print upright as exported.
C2_housing_half_2_x1	1	C2 housing half (carries fins 2 and 3, seam counterbores)	Mirror of C1 with pegs.
P_fin_x4	4	Fin (cam-driven panel, 90° stop)	Identical x4; hinge bore 4.5 mm; inward catch lug (plate + wall) on the top of the shell.
L_catch_cap_x4	4	Catch cap (open-top clip with screw boss)	Identical x4; open-top clip with an upward hook finger; print boss down so the 2.6 mm pilot is vertical.
R_rotor_x1	1	Rotor: shaft, 4 cams, crown arms, knob	Four 3.4 mm vertical holes in the crown arms.
U_output_hub_x1	1	Cartridge output hub	
Q_thrust_collar_x1	1	Lower thrust collar	Stays bonded (2.5 mm tall ring).
X_double_d_adapter_x1	1	Double-D adapter	
T_winding_cap_x1	1	Winding and latch cap	
D_cartridge_dock_x1	1	Cartridge dock (3 ring pads, 4 feet)	Feet have 3.4 mm radial holes.

File	Qty	Part	Notes
N_bayonet_collar_x1	1	Bayonet retaining collar	
I_locking_pin_x1	1	Quarter-turn locking pin	
M_casing_x1	1	Cartridge casing (3 lid bosses)	
F_lid_x1	1	Cartridge lid (3 screw ears)	
A_pushrod_x1	1	Short side pushrod	
B_release_yoke_x1	1	Upward release yoke	
S_servo_cradle_x1	1	SG90 cradle (4 pads, 2.6 mm pilots)	
G_servo_keeper_x1	1	Servo keeper	Stays bonded (1.5 mm plate).
J_horn_saddle_x1	1	Double-arm horn saddle	Measure your horn first (as in your model).
samples/S_unloaded_spiral_*_x1	1 each	Unloaded spiral springs 1.6 / 2.0 / 2.4 mm beam	Print references from your Part Studio 24, unchanged.
samples/W_unloaded_return_leaf_x1	1	Return leaf (1 mm installation preload)	Root stays bonded.

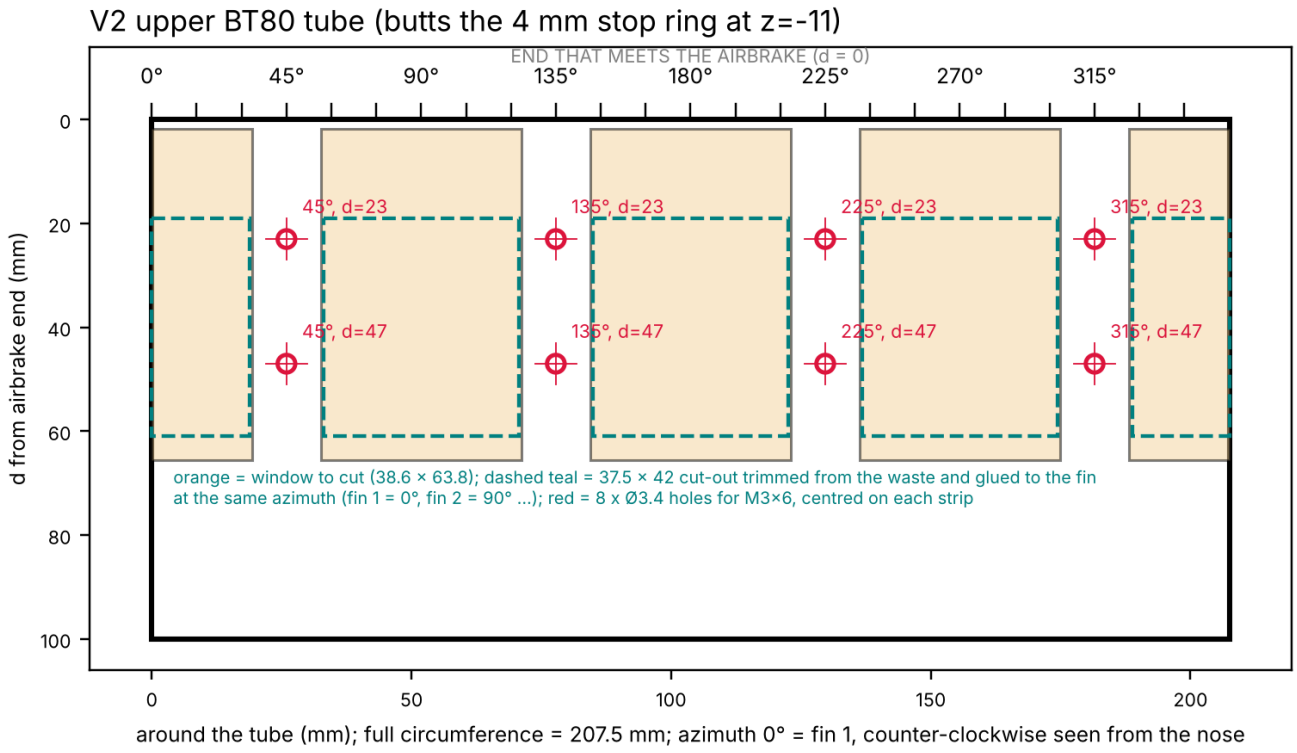
4 Hardware and tools

Ref in CAD	Fastener	Qty	Joint	Driver
HW seam screw 1-4	M3 x 8 socket-head cap (ISO 4762)	4	C2 counterbore → thread into C1 across the seam; 2 per seam at z=8 and z=44	2.5 mm hex
HW L1-L4 cap screw	M3 x 8 button head (ISO 7380)	4	Rotor arm → catch cap boss (vertical)	2.0 mm hex
HW H1-H4 hinge screw	M4 x 25 button head (ISO 7380)	4	Through -Y ear and fin barrel, threads into +Y ear (replaces printed pin H and cap K)	2.5 mm hex
HW top-tube support screw 1-8	M3 x 5 button head	8	Upper BT80 tube → thick supports between the fins (z=12 and z=36)	2.0 mm hex
HW dock ring screw 1-3	M3 x 10 button head	3	Lower BT80 tube → housing sleeve → dock ring pad (z=-26.5)	2.0 mm hex
HW cradle screw 1-4	M3 x 10 button head	4	Lower BT80 tube → dock foot → cradle pad (z=-67.2)	2.0 mm hex
HW lid screw 1-3	M3 x 10 button head	3	Lid ear → casing boss (vertical)	2.0 mm hex

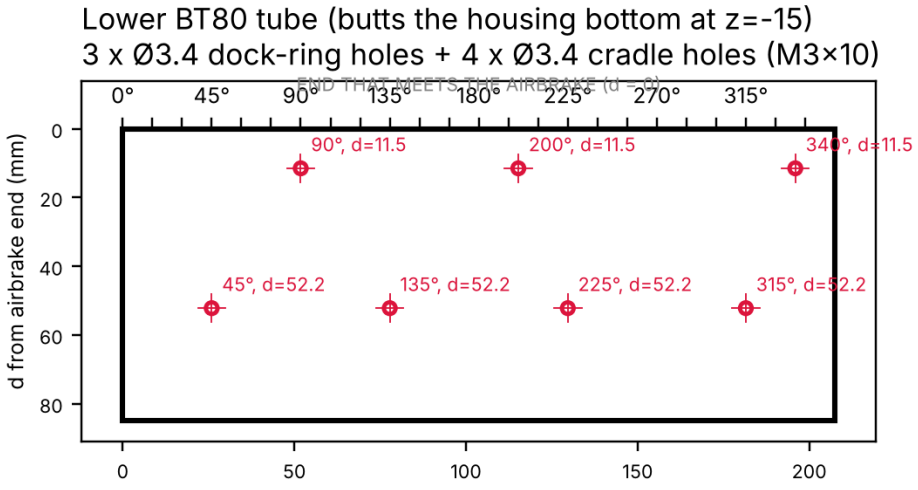
Totals: M3x5 × 8, M3x8 × 8 (4 socket + 4 button), M3x10 × 10, M4x25 × 4 = 30 screws. Not printed or screwed: 1 SG90 servo with horn, 2 BT80 airframe tubes, epoxy for Q, G, the return-leaf root and the horn saddle J.

5 Prepare the upper tube (slits) and the cut-outs

Cut the four windows (38.6 × 63.8 mm, centred on 0/90/180/270°, starting 1.8 mm from the end that meets the airbrake). From each piece of waste trim a **37.5 × 42 mm** cut-out (the dashed outline), edges square, ~0.3 mm sanded off the corners. That leaves about 0.6 mm of kerf at each side and about 4.6 mm above, and about 17 mm of open window below the cut-out so the fin can swing down and out. Drill the eight 3.4 mm screw holes (the strips between the windows). The lower tube is drilled exactly as in V1.

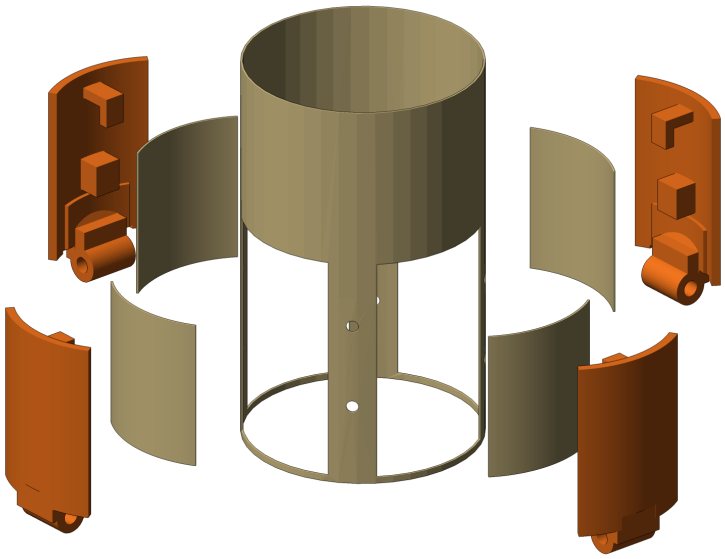


Upper tube template (unrolled): windows to cut, the cut-out piece trimmed from each waste, and the eight holes.



Lower tube template, same as V1.

Tube	Feature	d (mm)	Azimuth	Note
Upper	4 windows	1.8 → 65.6	0, 90, 180, 270° ±33.5°	38.6 wide at the OD
Upper	4 cut-outs	19 → 61	same centres, ±32.5°	37.5 × 42 mm, glued to the fins
Upper	8 holes (M3x5)	23 and 47	45, 135, 225, 315° (both rows)	centred on each strip
Lower	3 holes (M3x10)	11.5	90, 200, 340°	ring screws
Lower	4 holes (M3x10)	52.2	45, 135, 225, 315°	cradle screws

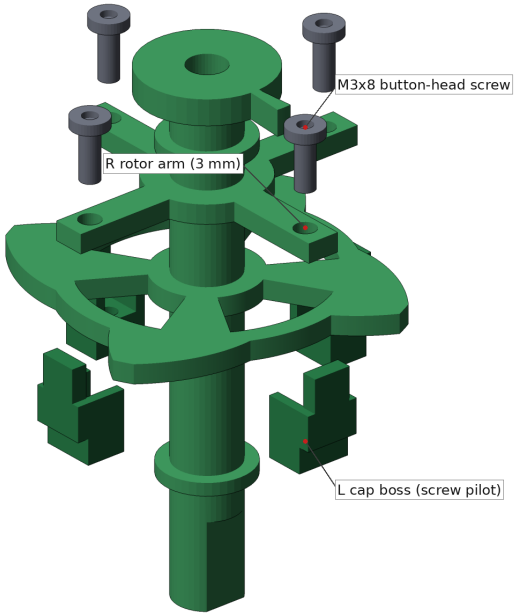


The upper tube with its four slits, the four cut-out pieces (tan) and the fins (orange) they are glued to.

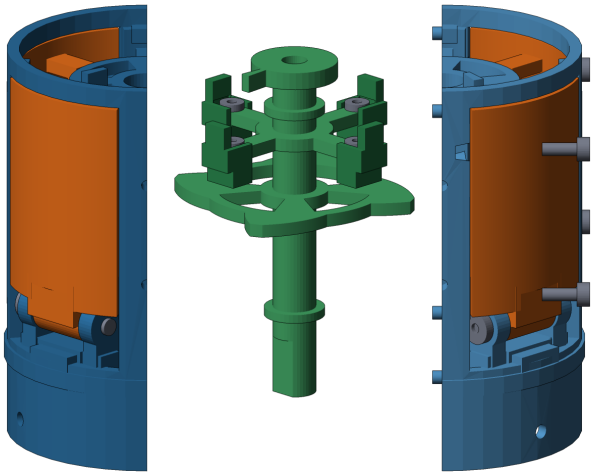
6 Assembly sequence

The order is the same as the V1 guide (rotor group, fins on the halves, close the housing, cartridge and cradle, module into the lower tube). Differences:

- **Catch caps:** the finger points up. Offer each cap from below with the finger on the **outer** side of the arm tip; the screw is M3x8 at R20.4 (a clip with the arm in the open-top socket, boss under it).
- **Latch set-up:** with the rotor at 0° each finger must stand outside the lug wall of its fin. Swing a fin out by hand: it should stop after about 1 mm. Turn the rotor to the release angle: the fin must then swing free.
- **Seam screws:** the lower pair sit on ribs at z=8 as in V1; the upper pair (z=44) are M3x8 in the thick strips at R29.6, counterbore in the strip face at the window edge of the C2 side. Swing fin 2 (for the 45° seam) and fin 3 (225° seam) open to reach them.
- **Fins** have R29.9–32.3 shells with the lug at the top; the cut-out pieces are added in step 7.
- **Top ring** (z 54.6–58.8) replaces the 18 mm upper sleeve; the rotor knob (z 52–56) sits in it and the top bearing is at z 46–51.



Caps offered up under the arms (fingers up), screws above.



V2 housing halves apart: the thick strips and the short top ring replace the upper sleeve.

Step 5 – Module and lower tube

Exactly as V1 step 5 (3 ring screws and 4 cradle screws M3x10 through the lower tube).

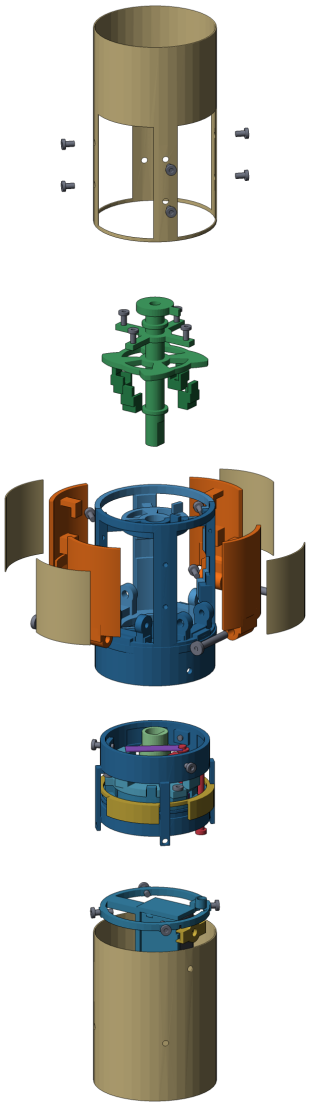
Step 7 – Upper tube and the glued cut-outs

Close all four fins and lock the latch (rotor at 0°). Slide the airbrake head into the **upper tube from its cut end** (the tube butts the 4 mm stop ring at z=–11) and rotate it so the windows sit over the fins. Fit the 8 M3x5 button-head screws through the tube into the strips: with the tube on you should see all eight pilot holes through the tube holes.

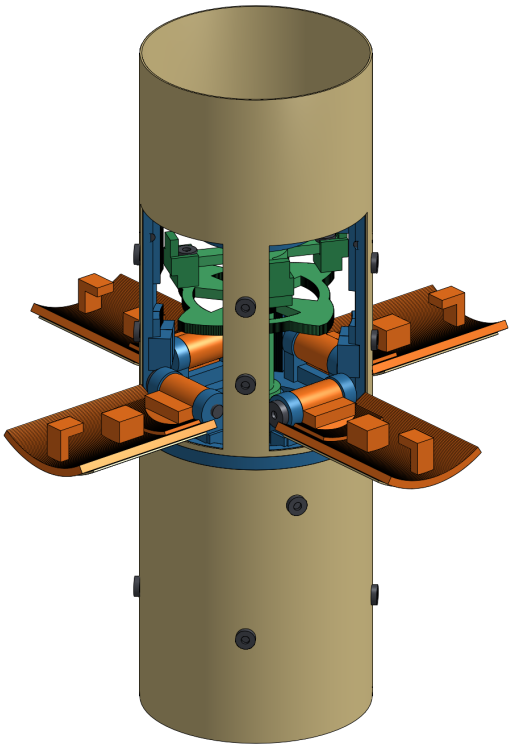
Now glue each cut-out onto its fin: wipe the fin's outer face, put a thin layer of epoxy or CA gel on the fin (0.2 mm gap = glue line), place the cut-out in the window, and hold it with masking tape across the window until cured. **Keep glue off the window edges and the surrounding tube** (a strip of tape or wax on the tube edges) or the fin will be glued shut. After curing, cut any glue bridge and check that the fin lifts out of its window.

Step 8 – Function check

- Fins open to 90° carrying their cut-outs: the cut-out lower edge swings down through the open lower part of the window and clears the tube (checked at 15/40/65/90°, no interference).
- Latch: rotor 0° holds the fins; at the release angle the fins are free; the cams then kick them (rotor up to 65°).
- Eight tube screws and seven lower-tube screws snug (15 screws touch the tubes in total).



V2 exploded: upper tube with its eight screws, rotor, fins with hinge screws and glued cut-outs, housing with supports, cartridge, cradle, lower tube.



V2-B: fins at 90° carrying their cut-outs, rotor at 65° (Onshape render).

7 Checks run on this model

Check	Result
V2-A closed	64 parts, all watertight; printed-part interference: none; screw/thread engagement pairs: 36 (intended).
V2-B open (90°, 65°, 4.5 mm)	64 parts, all watertight; printed-part interference: none; screw/thread engagement pairs: 36 (intended).
Latch hold / release / cam timing	blocked from 2° fin opening at rotor 0°; free from rotor 16°; cams touch by 26° (not at 22°)
Cut-outs TP1–4 vs housing, fins and tube at 15/40/65/90°	No overlap (they start at z=8 so the lower edge clears the stop block)
Diameters	Fin OD 64.6; tube ID 65.0; tube OD 66.04; support OD 64.6; stop ring OD 66.04 (all read from the CAD)
Seam screw access	Straight 2.5 mm hex key reaches all four heads with fins 2 and 3 open (ray test)
Assembly mates	28 features (adds a group per fin with its cut-out): all solve OK

Open points for you: (1) The new catch is a different mechanism from your tab-and-hook; its strength and the 16° release angle have not been tested on a print. (2) The tube keeps only a 1.8 mm ring below the windows: epoxy it to the stop ring and rely on the eight screws. (3) Cam peak was reduced 29 → 28.6 mm for pillar clearance, which trims the kick slightly. (4) Mate motion not scripted: drag a fin in ASM-V2 to confirm.